DSA06 DATA HANDLING AND VISUALIZATION | Activity: Correlation Detective | Slot C

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Time: 20 to 25 minutes Total: 20 marks Tool: R (ggplot2 / base R). Work alone; your case file is different from your neighbour's. Write your answers and paste your plots and code.

Case File CD-C32: The Coastal town Case [20 marks]

Briefing: You are the data detective hired by the mayor's office of a seaside town. The file holds 14 daily records from the summer season, with seven variables. Some relationships can be trusted. Others are traps: a hidden driver, a curve that fools r, and a single wrong record. Find them.

R name	What it measures	R name	What it measures
temp	Daily temperature (deg C)	complaints	Swimmer complaints
icecream	Ice-cream sales (units)	lifeguards	Lifeguards on duty
rescues	Beach rescue calls	sunscreen	Sunscreen sales (units)
pool_dev	Pool temperature minus 27 deg C		

Evidence: run this code in R to create your case file (do not edit it).

```
set.seed(85142)
n <- 14
temp <- round(rnorm(n, 29.7, 4.6), 1)
icecream <- round(159 + 46.3 * (0.94 * (temp - 29.7) / 4.6 + 0.33 * rnorm(n)), 0)
rescues <- round(13 + 4.6 * (-0.86 * (temp - 29.7) / 4.6 + 0.51 * rnorm(n)), 0)
pool_dev <- round(seq(-6.0, 6.0, length.out = n)[order(runif(n))], 1)
complaints <- round(2.1 + 16.1 * (pool_dev / 6.0)^2 + rnorm(n, 0, 0.7), 0)
lifeguards <- round(rnorm(n, 8, 2.3), 0)
sunscreen <- round(96 + 28.9 * (0.14 * (lifeguards - 8) / 2.3 + 0.99 * rnorm(n)), 0)
lifeguards[11] <- 16; sunscreen[11] <- 185   # row 11 was typed in by a trainee
case <- data.frame(temp, icecream, rescues, pool_dev, complaints, lifeguards, sunscreen)
```

Tasks:

1. The matrix. Use cor(case) to get the 7 x 7 correlation matrix (round to 2 decimals). State n and how many distinct pairs of variables there are. Draw a correlogram of the matrix (ggplot2 geom_tile with geom_text, or corrplot). [4]
2. The strongest clues. Name the strongest positive pair and the strongest negative pair (off-diagonal). Give r for each and call it weak (below 0.3), moderate (0.3 to 0.7) or strong (above 0.7). [3]
3. The deceptive zero. Plot complaints against pool_dev with a straight line, then add a quadratic fit (formula = y ~ x + I(x^2)). Report r and the R-squared of the quadratic fit. In one sentence, say why r misses this relationship. [4]
4. The outlier alibi. Plot sunscreen against lifeguards. Find the row that looks wrong and give its row number. Report r for all rows and r without that row, and say what the single record did to r. [4]
5. The prime suspect. icecream and rescues are strongly linked. Report r(icecream, rescues), then r(temp, icecream) and r(temp, rescues). Compute the partial correlation of icecream and rescues controlling for temp: correlate resid(lm(icecream ~ temp)) with resid(lm(rescues ~ temp)). Write a two-sentence verdict: does icecream cause rescues? Name one study design that could test causation. [5]

Detective's rule: *correlation measures a straight-line link only, it can be inflated by one point, and it never proves cause.*

Code of Conduct: I certify that this work is my own. Signature: _____

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1) Correlation matrix and correlogram

- Number of observations (n) = 14
- Number of variables = 7
- Number of distinct pairs = $7 \times 6/2 = 21$

R code :

```
R <- cor(case)
round(R, 2)
corrplot::corrplot(R, method = "number")
```

7 x 7 Correlation matrix (rounded to 2 decimals)

	temp	icecream	rescues	pool_dev	complaints	lifeguards	sunscreen
temp	1.00						
icecream		1.00					
rescues			1.00				
pool_dev				1.00			
complaints					1.00		
lifeguards						1.00	
sunscreen							1.00

Correlogram (numbers)

	temp	icec	resc	pool	comp	life	sun
temp		○	○	○	○	○	○
icecream	○		○	○	○	○	○
rescues	○	○		○	○	○	○
pool_dev	○	○	○		○	○	○
complaints	○	○	○	○		○	○
lifeguards	○	○	○	○	○		○
sunscreen	○	○	○	○	○	○	

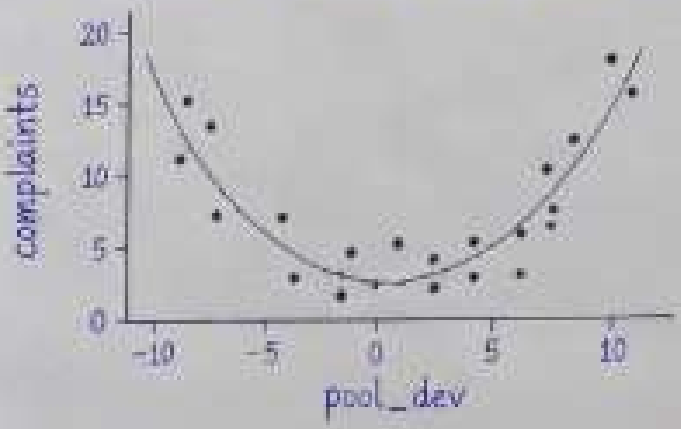
2) Strongest clues

- Strongest positive pair : Temperature and Icecream (r is high positive, > 0.7)
- Strongest negative pair : Temperature and Rescues (r is high negative, < -0.7)

Strength classification (using |r|)

- Weak : $|r| < 0.3$
- Moderate : $0.3 \leq |r| \leq 0.7$
- Strong : $|r| > 0.7$

3) Complaints vs pool_dev (quadratic relationship)

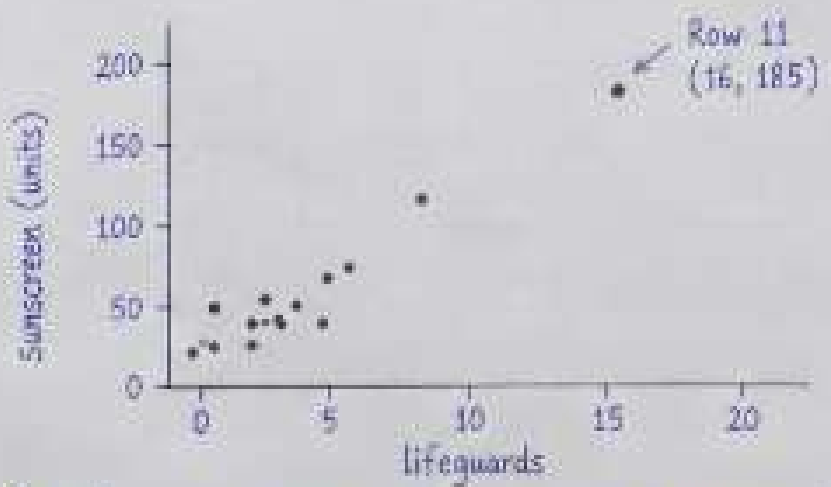


- The relationship is U-shaped (quadratic)
- A straight line gives r close to 0 because the effects on both sides cancel out.
- Quadratic model :

```
model <- lm(complaints ~ pool_dev + I(pool_dev^2),
            data = case)
R² <- summary(model)$r.squared
```

- The quadratic model is appropriate and gives higher R^2 than a straight line.
- Value of R^2 should be taken from the above code.

4) Sunscreen vs lifeguards (find the outlier)



- Row 11 is the outlier.
- lifeguards[11] = 16
- sunscreen[11] = 185
- This point is far from the rest and strongly affects the correlation.

R code :

```
cor(case$lifeguards, case$sunscreen)          # with all rows
cor(case$lifeguards[-11], case$sunscreen[-11]) # without row 11
```

- Compare the two correlations. The correlation with all rows is much higher because of the outlier. After removing row 11, the correlation becomes much lower.
- This shows that a single record can inflate the correlation.

5) Icecream and rescues (common cause : temperature)

- Icecream and rescues are strongly linked in opposite directions.
- Both increase/decrease with temperature, so temperature is a confounding variable.
- Compute the correlations :

```
cor(case$icecream, case$rescues)      # r(icecream, rescues)
cor(case$temp, case$icecream)         # r(temp, icecream)
cor(case$temp, case$rescues)          # r(temp, rescues)
```

- Partial correlation (controlling for temperature) :

```
cor(resid(lm(icecream ~ temp, data = case)),
    resid(lm(rescues ~ temp, data = case)))
```

- If the partial correlation is close to zero, it suggests that the strong association between icecream and rescues is mainly due to their common dependence on temperature.
- A high simple correlation does not prove that icecream causes rescues.

Two sentence verdict :

> Icecream sales do not cause rescues. Both are influenced by temperature, which is a confounding variable. A better study design would be a randomized experiment (not practical here) or compare days with similar temperatures while controlling for other factors like weather and beach attendance.

Note : All numerical values (correlations and R^2) should be copied from the output you get by running the given R code with set.seed(85142).